Logic Gates :

- A digital logic gate is a circuit that performs a logic function on a no. of input binary signals. The logic gate is the building block from which many kinds of logic circuits can be constructed. The signals at the i/p and o/p terminals of gate are either at High level voltage or at a low level.
- The logic functions are defined by a Truthtable.
- The Truth Table represent all the possible combinations of the i/p variables applied to a gate and its o/p variable. Each variable can only be at either '0' (logic 0 level) or 1 (logic 1 level).

→ The Basic Logic Gates are

1) AND gate
2) OR gate
3) NOT gate

→ Derived Logic Gates are .

1. NAND gate
2. NOR gate
3. EX-OR gate (XOR gate)
4. EX-NOR gate (XNOR gate).

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→ **AND Gate**: It is an electronic circuit that performs logical multiplication.

- The o/p of AND gate will be 'high' only when all of the inputs are 'high'.
- The AND gate is composed of two or more i/ps and a single o/p.
- The logic symbol of the AND gate with two i/ps A and B and the o/p Y is as shown below.

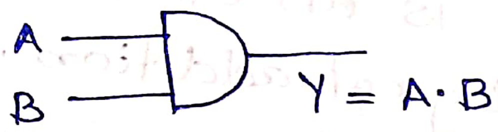
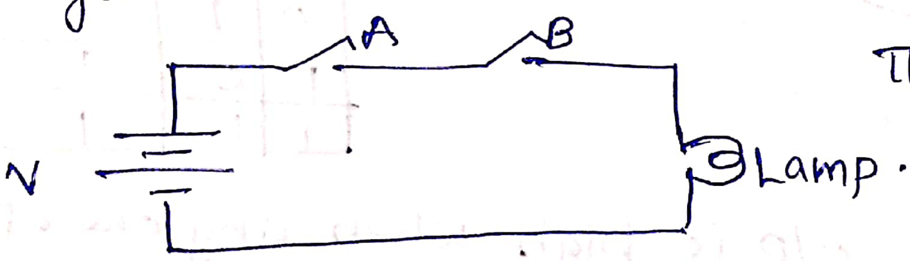


Fig: 2-input AND gate (symbol).

→ Truth Table (AND gate)

A	B	Y = A·B
0	0	0
0	1	0
1	0	0
1	1	1

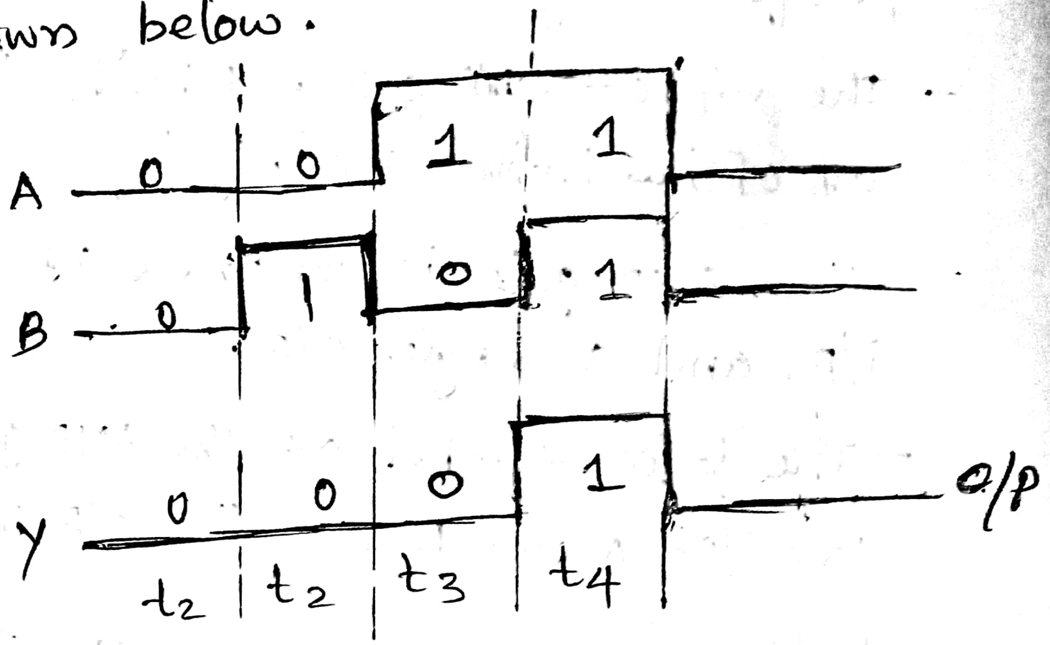
→ AND function representation by switches is as given below.



The lamp turns ON only when both switches A and B are closed.

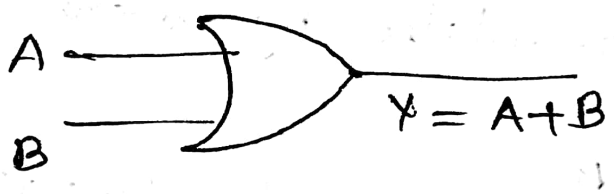
when switch is open, it is logical '0'.
when switch is closed, it is logical '1'.
logic '0' represents a low level voltage.
logic '1' represents a high level voltage.

The i/p and o/p pulse waveforms of AND gate is shown below.



→ **OR gate** : It is an electronic ckt that performs logical addition.

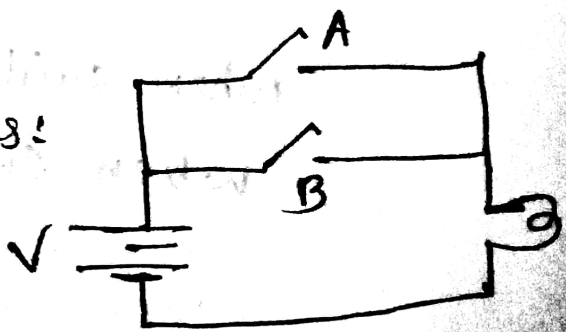
- It has two or more i/p and one o/p.
- The logical symbol for a two-i/ps OR gate is shown below.

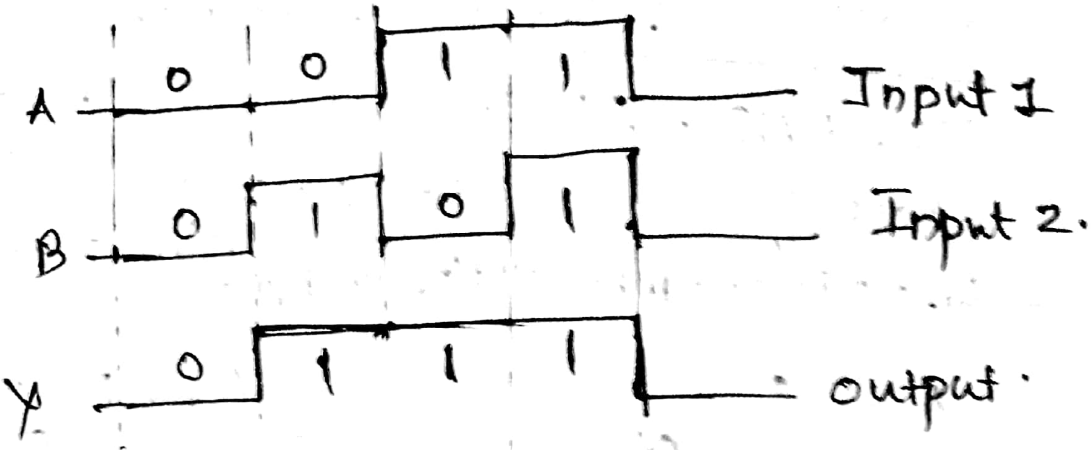


A	B	Y = A+B
0	0	0
0	1	1
1	0	1
1	1	1

- The o/p is high when any one of the i/ps is high.

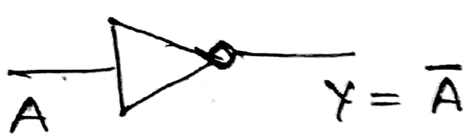
- OR function by switches:
- Bulb turns on when either A or B is closed.





→ <u>NOT gate</u> : It performs a basic logic function called Inversion or complementation.

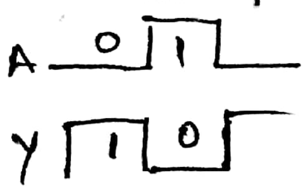
- It has only one i/p and one o/p.
- When high level (1) is applied at the i/p, a low level (0) is obtained at the o/p and vice versa.
- The o/p will be the complement of the i/p
- Logic symbol is shown below

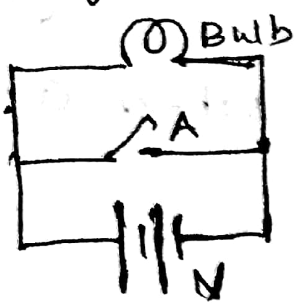


A	$Y=\overline{A}$
0	1
1	0

- Electrical equivalent is given by the fig below.

- waveform

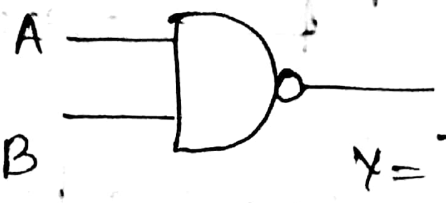




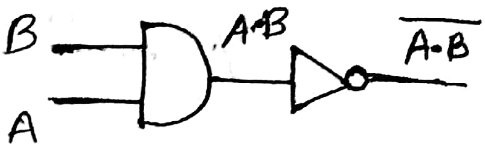
Bulb turn ON only when the switch is open (ie A = 0).

⇒ NAND gate :, It is a combination of NOT-AND. It performs the complemented AND function, (ie; AND gate followed by an inverter).

- The o/p is Low when all i/ps are high.



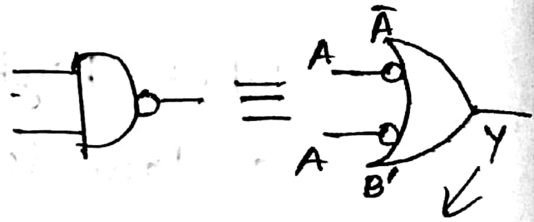
$Y = \overline{A \cdot B}$



Equivalent ckt.

A	B	$Y = \overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

$\overline{A \cdot B} = \bar{A} + \bar{B}$



$Y = \bar{A} + \bar{B}$

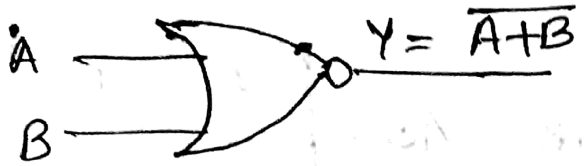
(equivalent negative OR gate)

Bubbled OR gate.

⇒ NOR gate :

- It is a combination of NOT-OR.
- It performs the complemented OR function. (ie; OR gate followed by an inverter)
- The o/p is low when any one i/p is high.

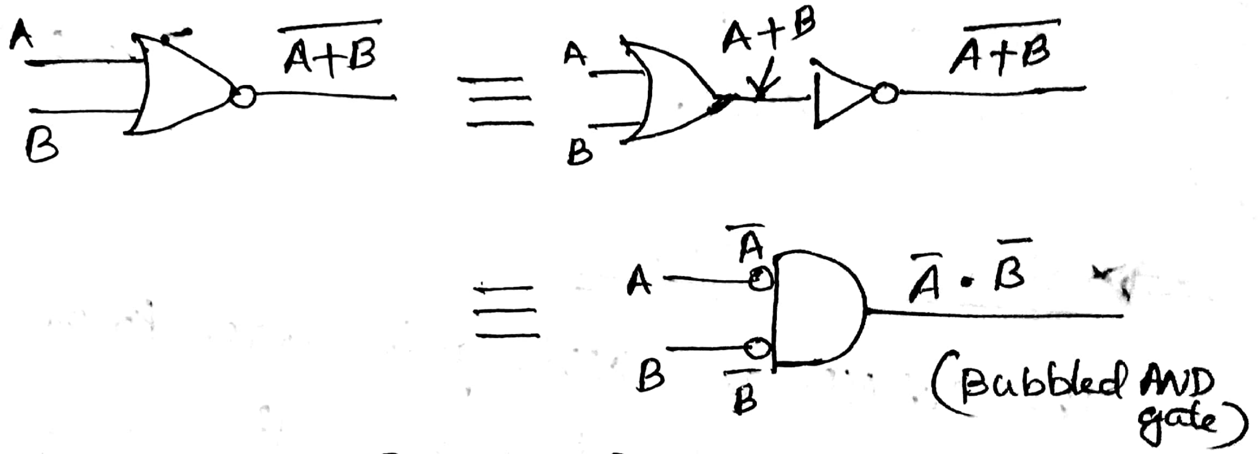
HW ⟶ i/p and o/p waveform (HW). for NOR & NAND gate



A	B	$Y = \overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

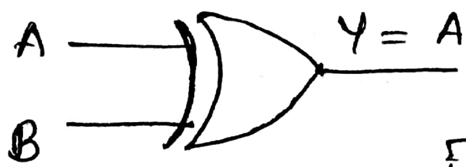
$$\overline{A+B} = \overline{A} \cdot \overline{B}$$

i.e; NOR gate is equivalent to negative AND gate.



⇒ EX-OR gate (XOR gate)

- The o/p of XOR gate is high only when the two i/ps are at opposite levels.

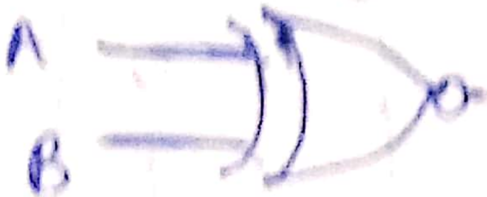


$$Y = A \oplus B = \overline{A}B + A\overline{B}$$

i/p and o/p waveforms. HW.

A	B	$Y = \overline{A}B + A\overline{B}$
0	0	0
0	1	1
1	0	1
1	1	0

⇒ EX-NOR gate :



$Y = \overline{A \oplus B} = A \odot B$

$Y = \bar{A}\bar{B} + AB.$

- o/p is the complement of the XOR function

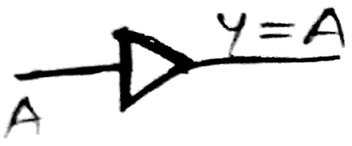
- o/p is high only when the two I/p are at the same level.

A	B	$Y = \overline{A \oplus B} = \bar{A}\bar{B} + AB$
0	0	1
0	1	0
1	0	0
1	1	1

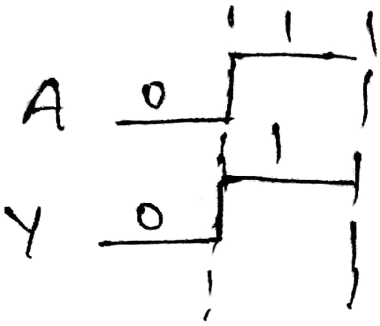
Truth Table.

I/p and o/p waveforms hw.

→ Buffer:

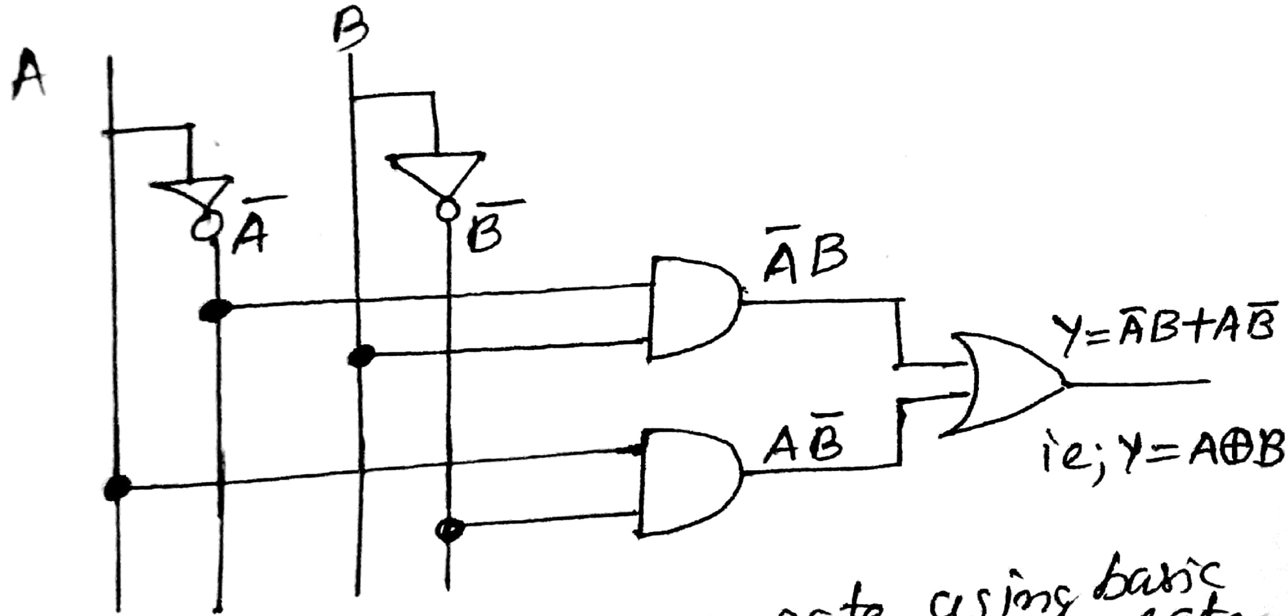


A	Y=A
0	0
1	1



Q: Implement an EX-OR gate using basic gates.

① $Y = A \oplus B = \bar{A}B + A\bar{B}$



② Implement an EX-NOR gate using basic gates.

$Y = A \odot B = \overline{A \oplus B} = \bar{A}\bar{B} + AB$